Results

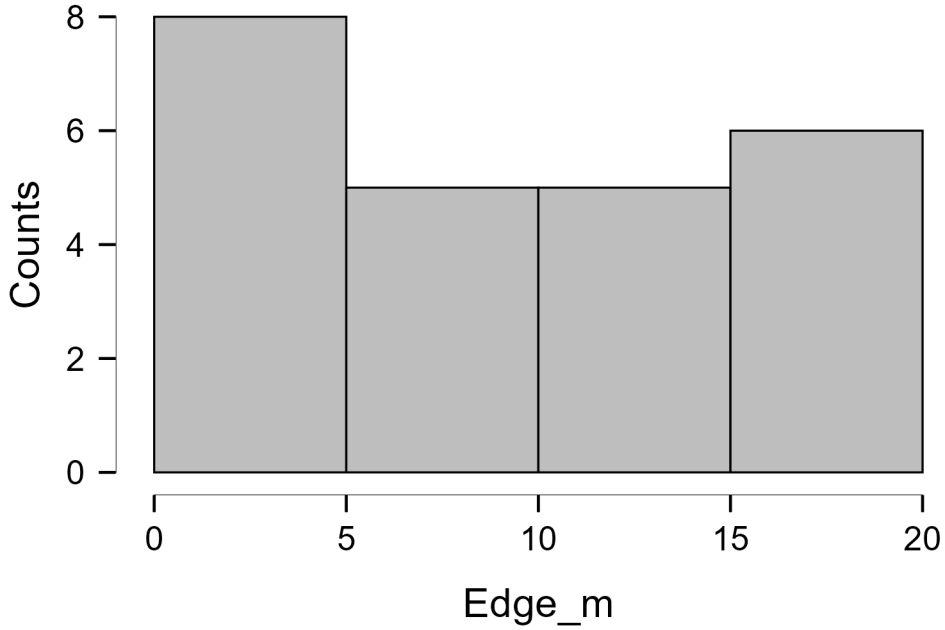
Descriptive Plots

Descriptive Statistics

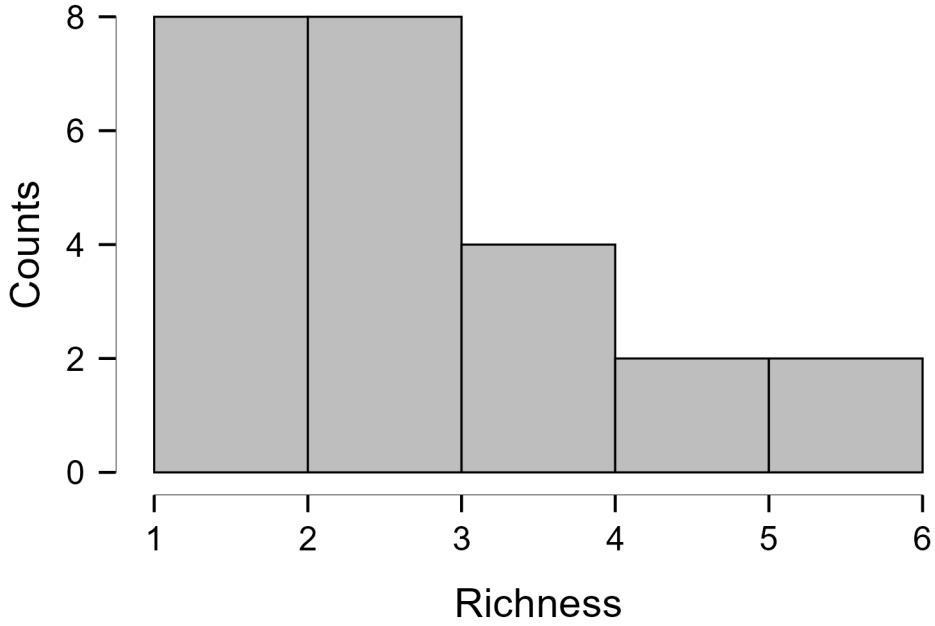
	Edge_m	Richness	log10_lux	Cover	Tree_m	Litter_cm
Valid	24	24	24	24	24	24
Mean (arithmetic)	9.604	3.125	0.7652	77.84	1.597	2.696
Std. Deviation	6.362	1.424	0.8130	10.81	0.9678	1.893
Minimum	0.000	1.000	−0.8539	52.71	0.08000	0.5000
Maximum	20.00	6.000	2.276	96.49	4.200	8.000

Distribution Plots

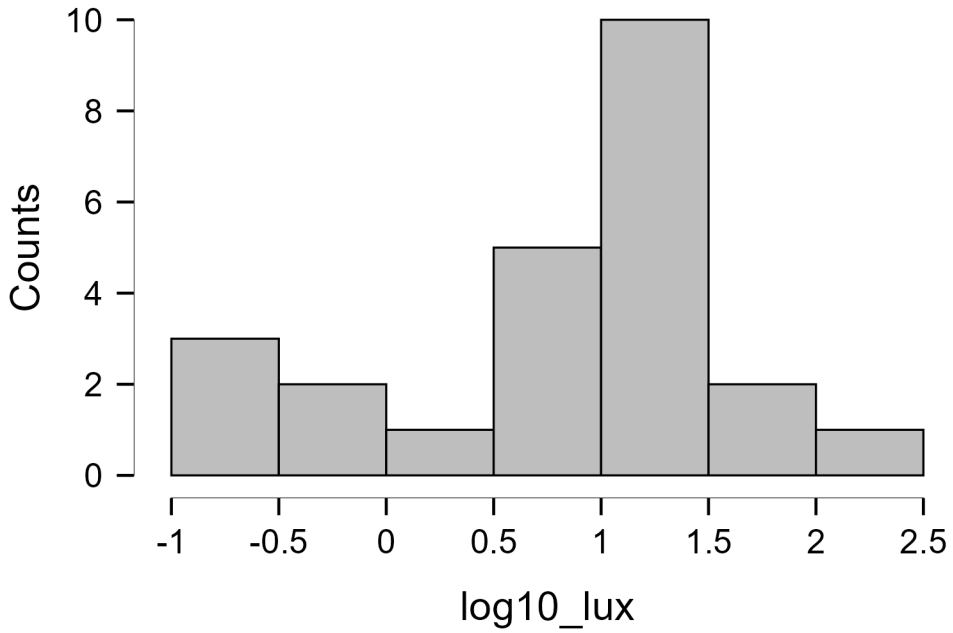
Edge_m



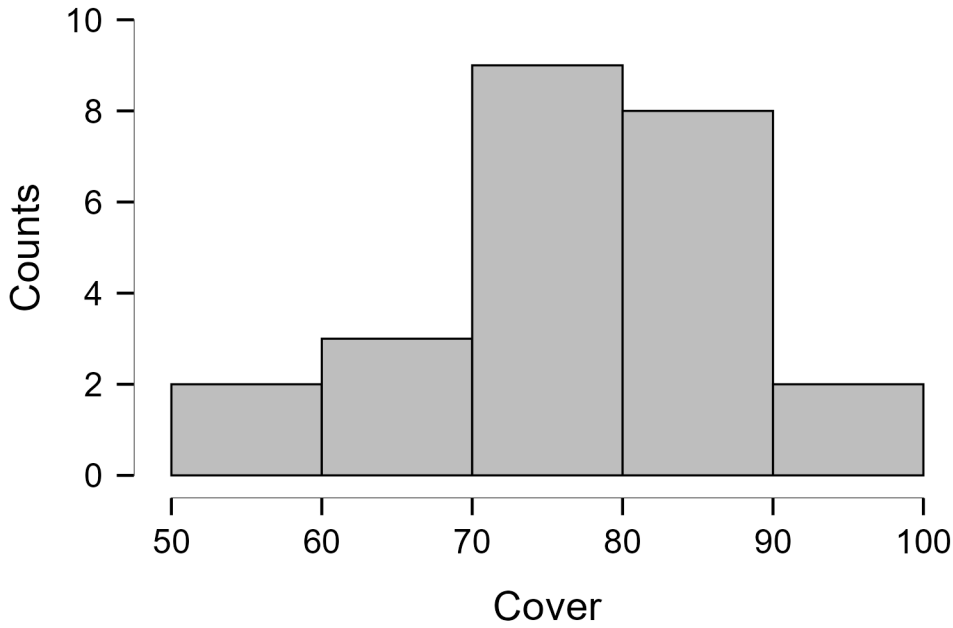
Richness



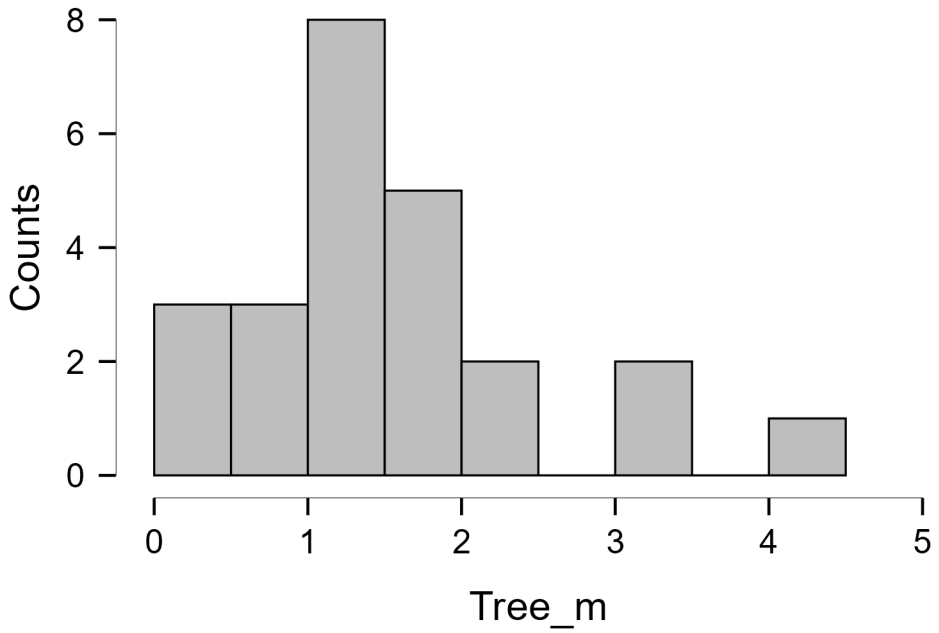
log10_lux



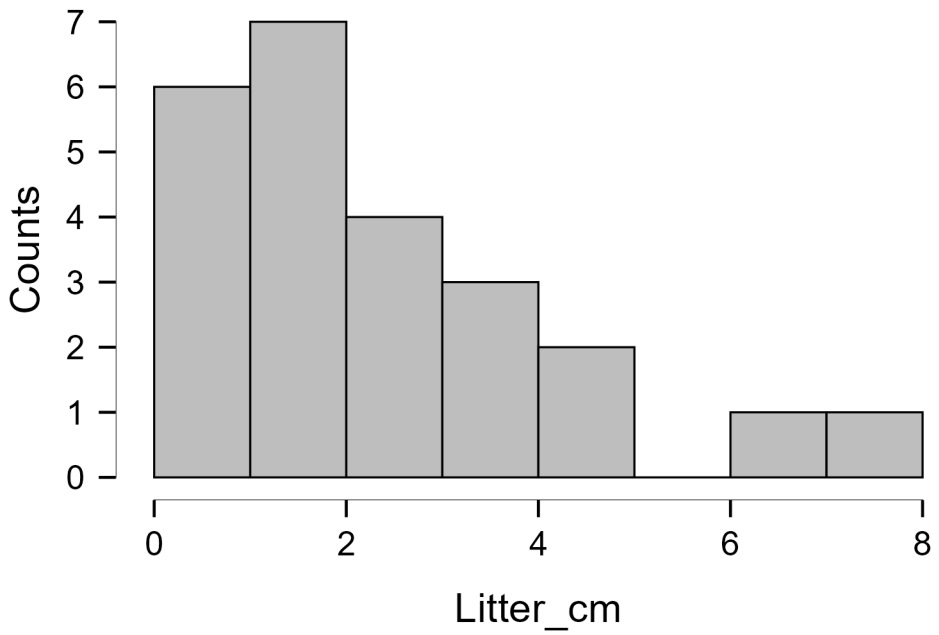
Cover



Tree_m



Litter_cm



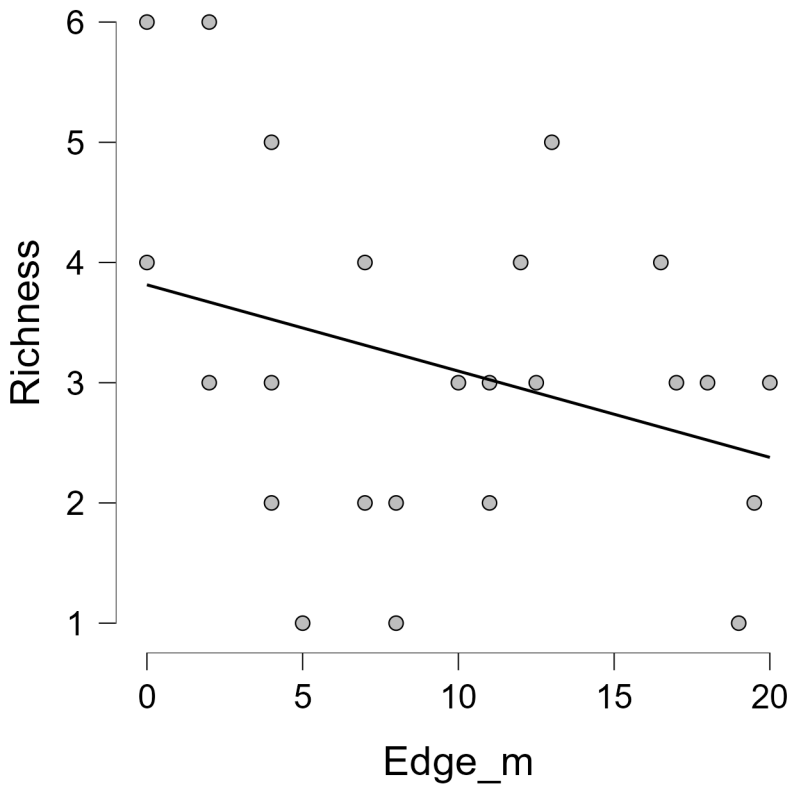
Variable Matrixes

Pearson's Correlations

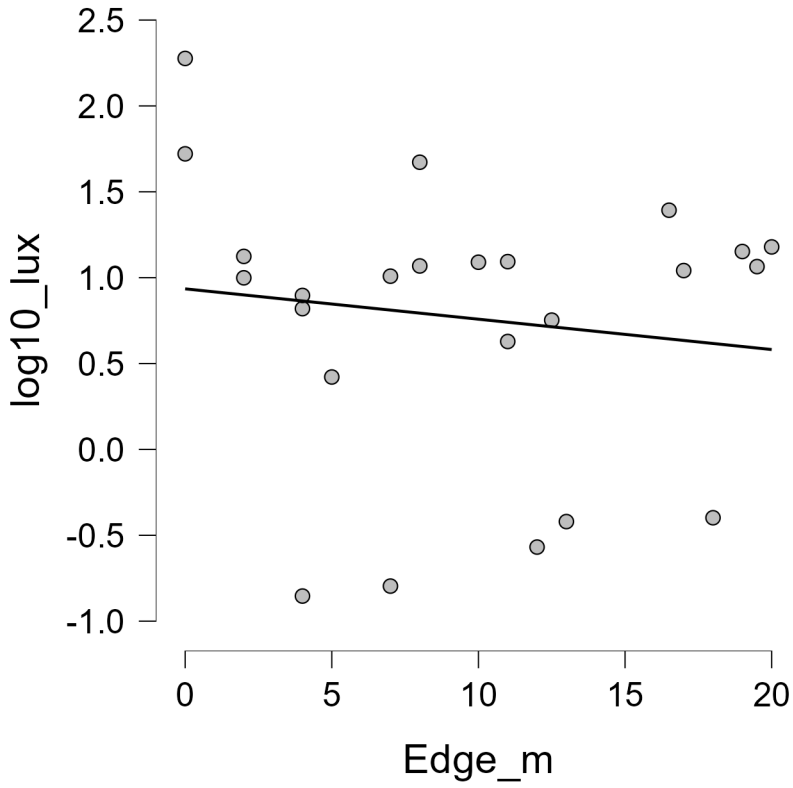
			Pearson's r	p
Edge_m	-	Richness	−0.3207	.127
Edge_m	-	log10_lux	−0.1383	.519
Edge_m	-	Cover	−0.08962	.677
Edge_m	-	Tree_m	−0.1265	.556
Edge_m	-	Litter_cm	−0.3274	.118
Richness	-	log10_lux	−0.1407	.512
Richness	-	Cover	−0.05201	.809
Richness	-	Tree_m	0.4190	.042
Richness	-	Litter_cm	0.3422	.102
log10_lux	-	Cover	−0.2665	.208
log10_lux	-	Tree_m	0.2650	.211
log10_lux	-	Litter_cm	0.2252	.290
Cover	-	Tree_m	−0.5225	.009
Cover	-	Litter_cm	0.1453	.498
Tree_m	-	Litter_cm	0.2132	.317

Scatter Plots

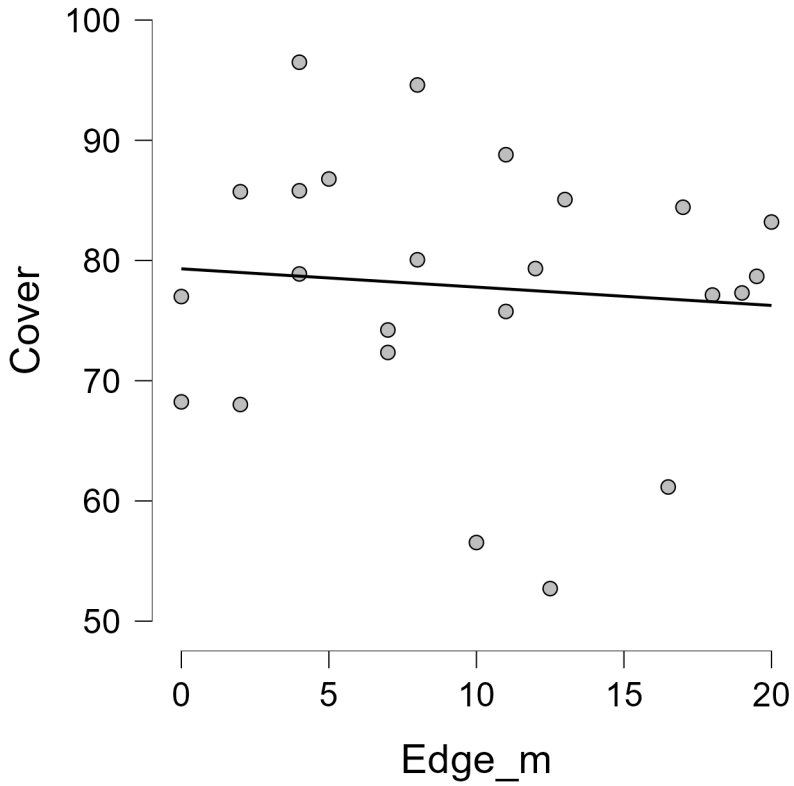
Edge_m vs. Richness



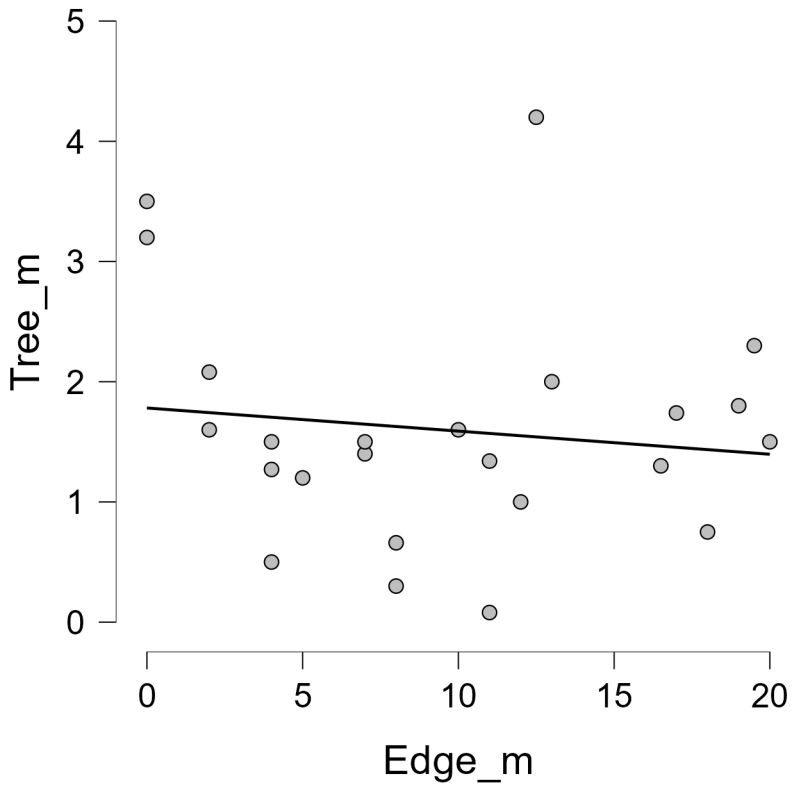
Edge_m vs. log10_lux



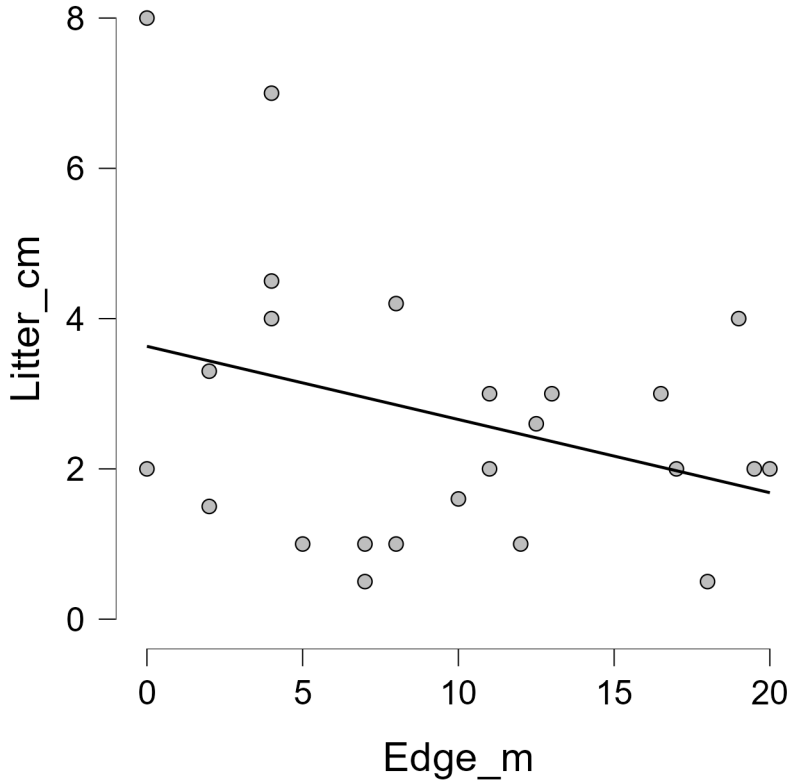
Edge_m vs. Cover



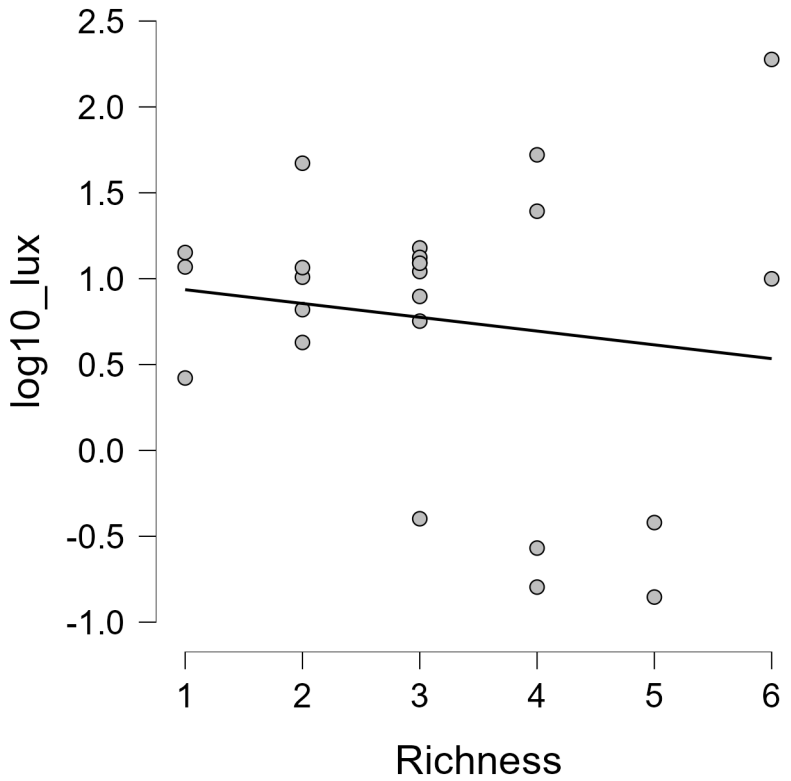
Edge_m vs. Tree_m



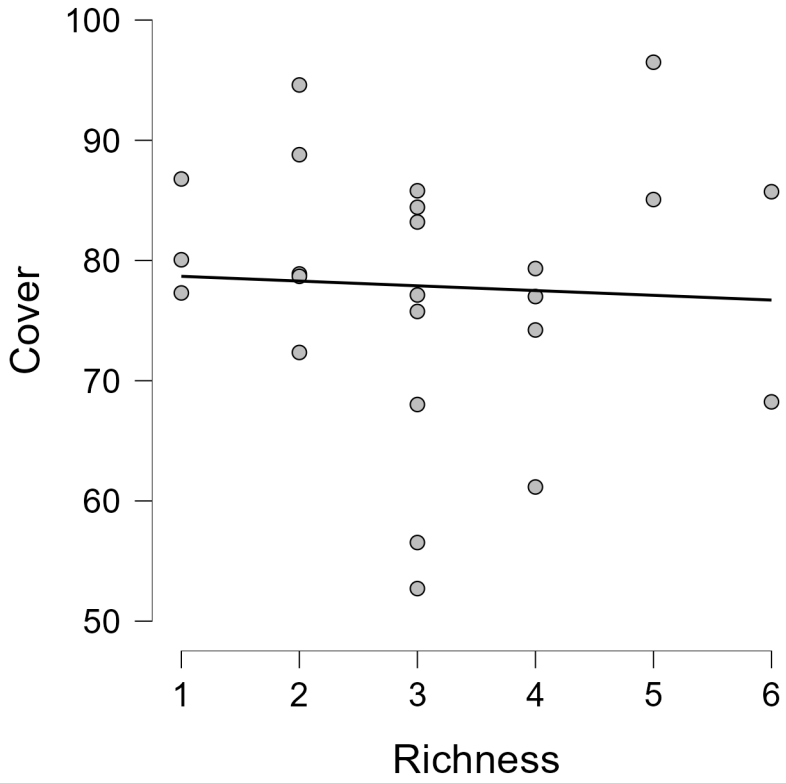
Edge_m vs. Litter_cm



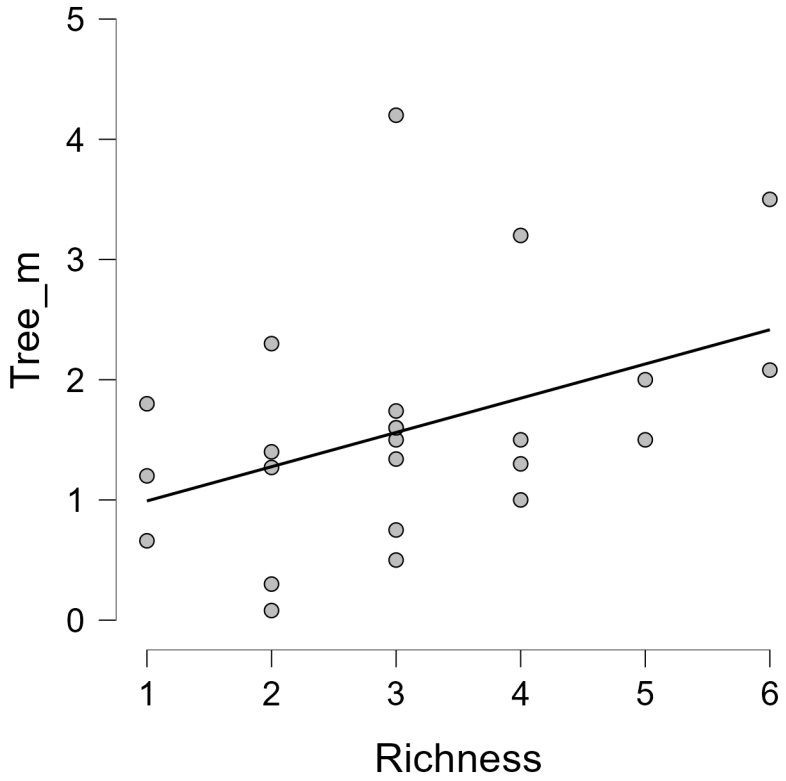
Richness vs. log10_lux



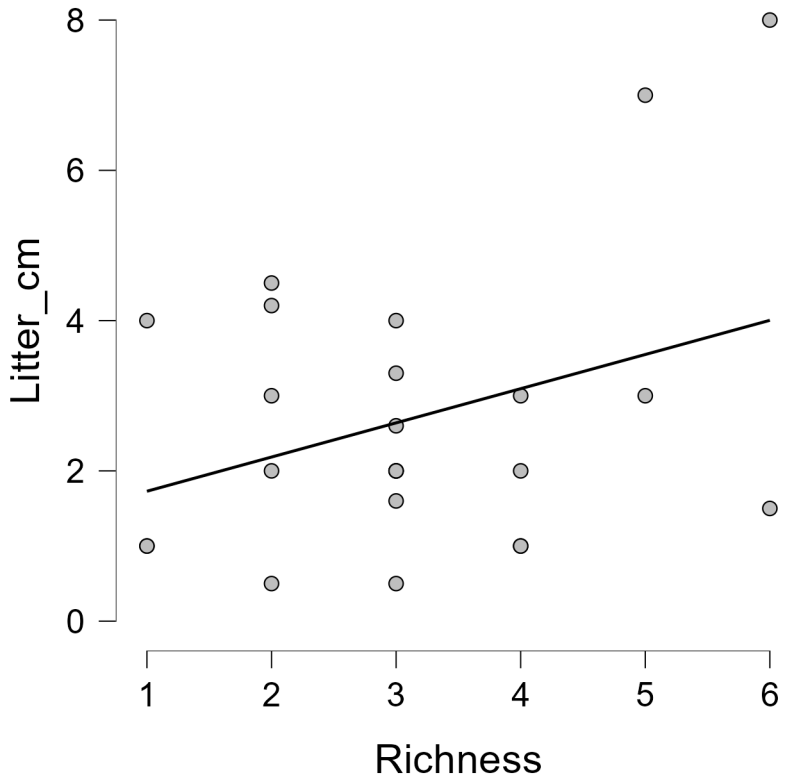
Richness vs. Cover



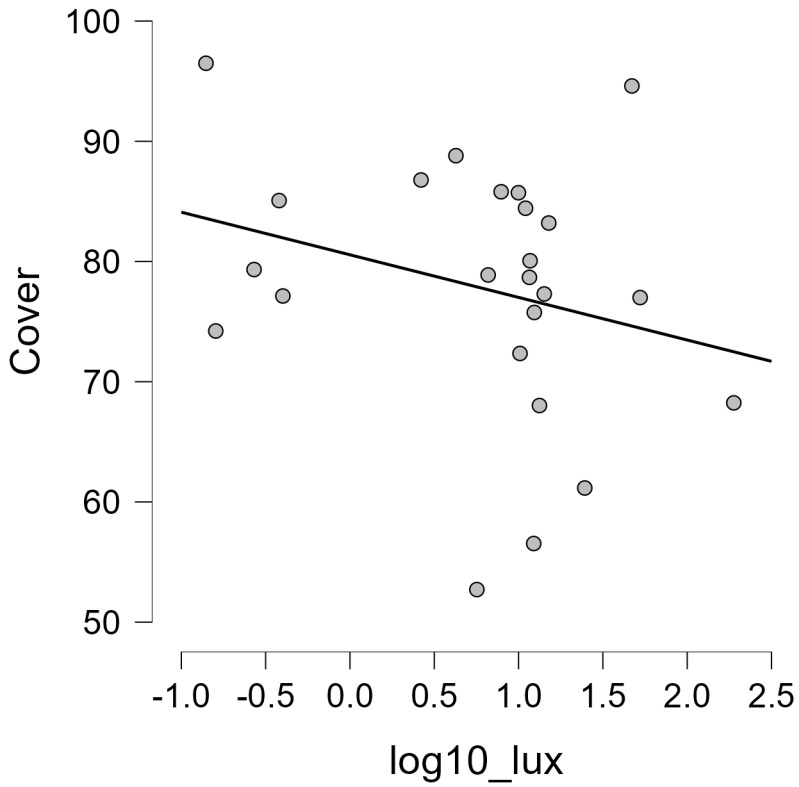
Richness vs. Tree_m



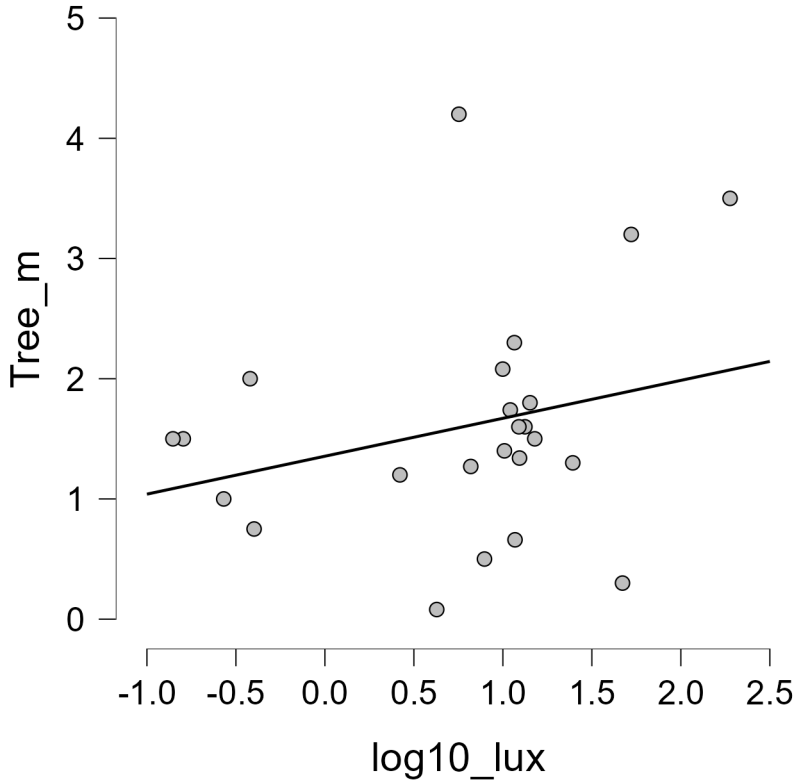
Richness vs. Litter_cm



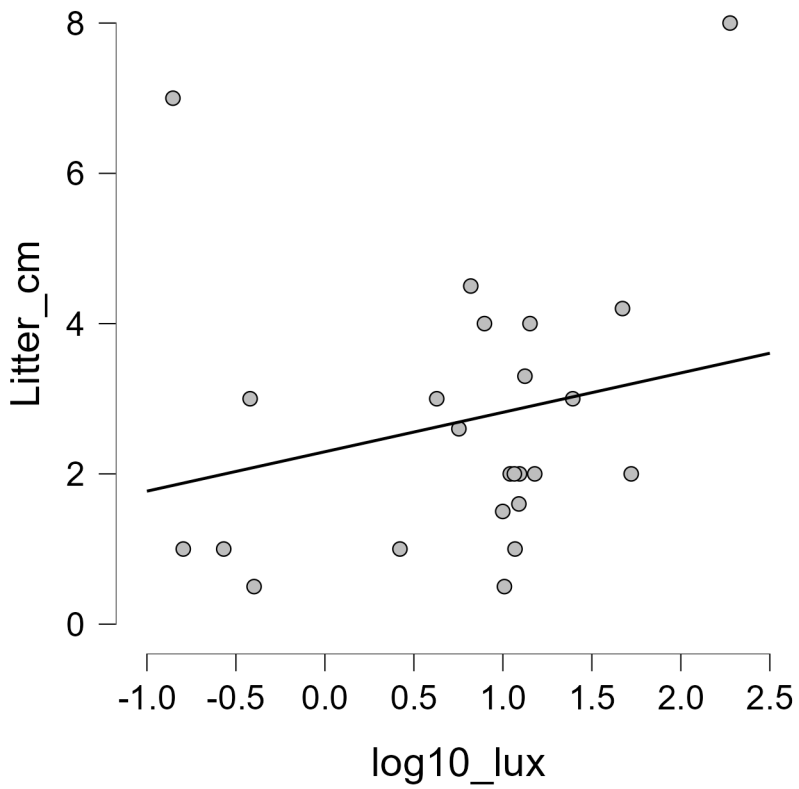
log10_lux vs. Cover



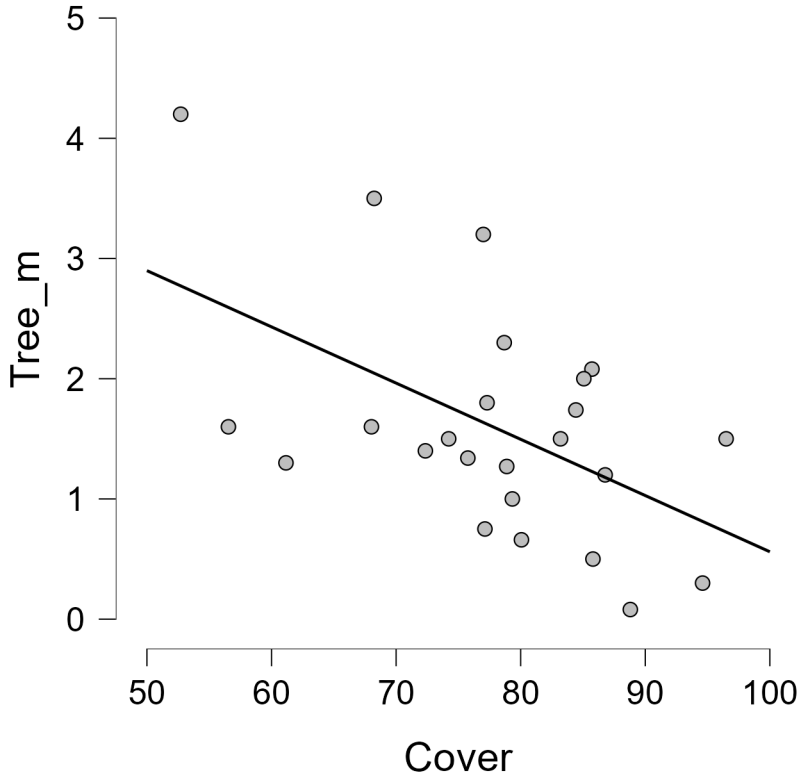
log10_lux vs. Tree_m



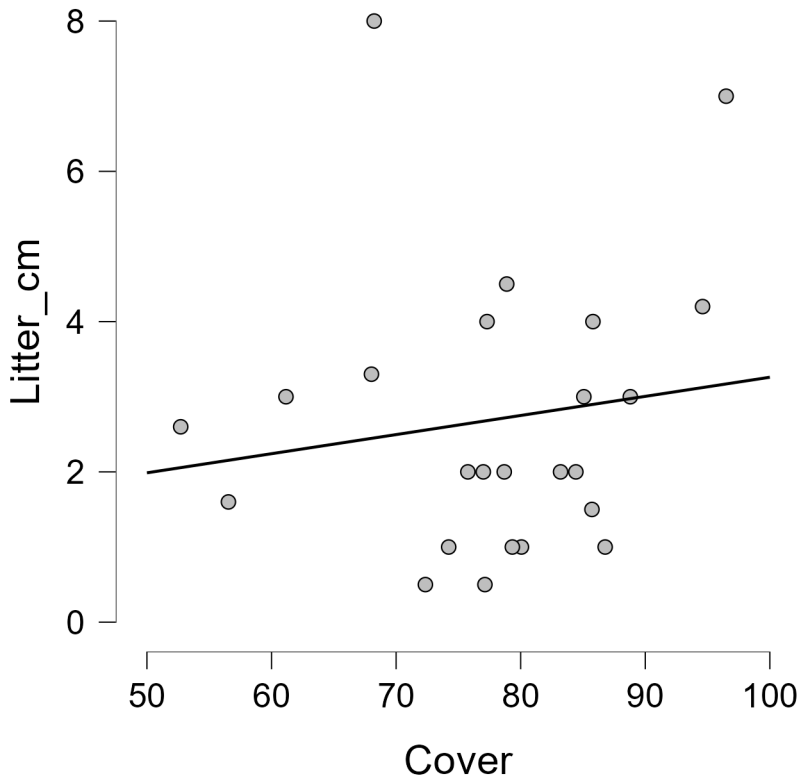
log10_lux vs. Litter_cm



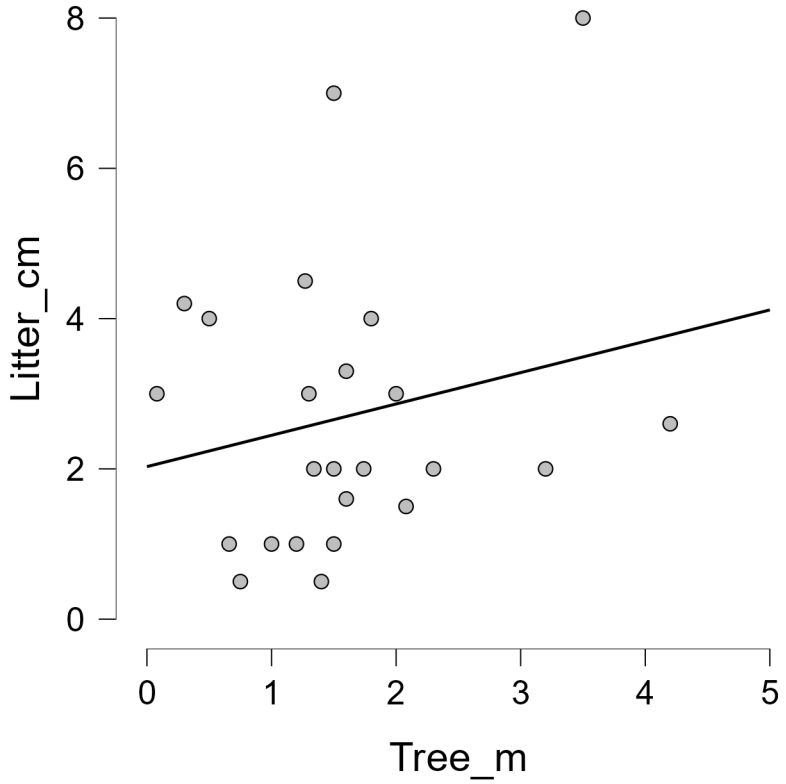
Cover vs. Tree_m



Cover vs. Litter_cm



Tree_m vs. Litter_cm



Linear Regression w/o Edge_m

Model Summary - Richness

Model	R	R^2	Adjusted R^2	RMSE	R^2 Change	df1	df2	p
M_0	0.000	0.000	0.000	1.424	0.000	0	23	
M_1	0.586	0.343	0.205	1.270	0.343	4	19	.079

Note. M_1 includes log10_lux, Cover, Litter_cm, Tree_m

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	15.99	4	3.998	2.479	.079
	Residual	30.63	19	1.612		
	Total	46.62	23			

Note. M_1 includes log10_lux, Cover, Litter_cm, Tree_m

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

							Collinearity Statistics	
Model		Unstandardized	Standard Error	Standardized	t	p	Tolerance	VIF
M_0	(Intercept)	3.125	0.2906		10.75	< .001		
M_1	(Intercept)	1.137	2.690		0.4227	.677		
	log10_lux	−0.5542	0.3524	−0.3164	−1.573	.132	0.854	1.171
	Cover	0.008815	0.03099	0.06694	0.2845	.779	0.625	1.601
	Litter_cm	0.2277	0.1551	0.3028	1.469	.158	0.813	1.229
	Tree_m	0.6963	0.3422	0.4733	2.034	.056	0.639	1.565

Coefficients Covariance Matrix

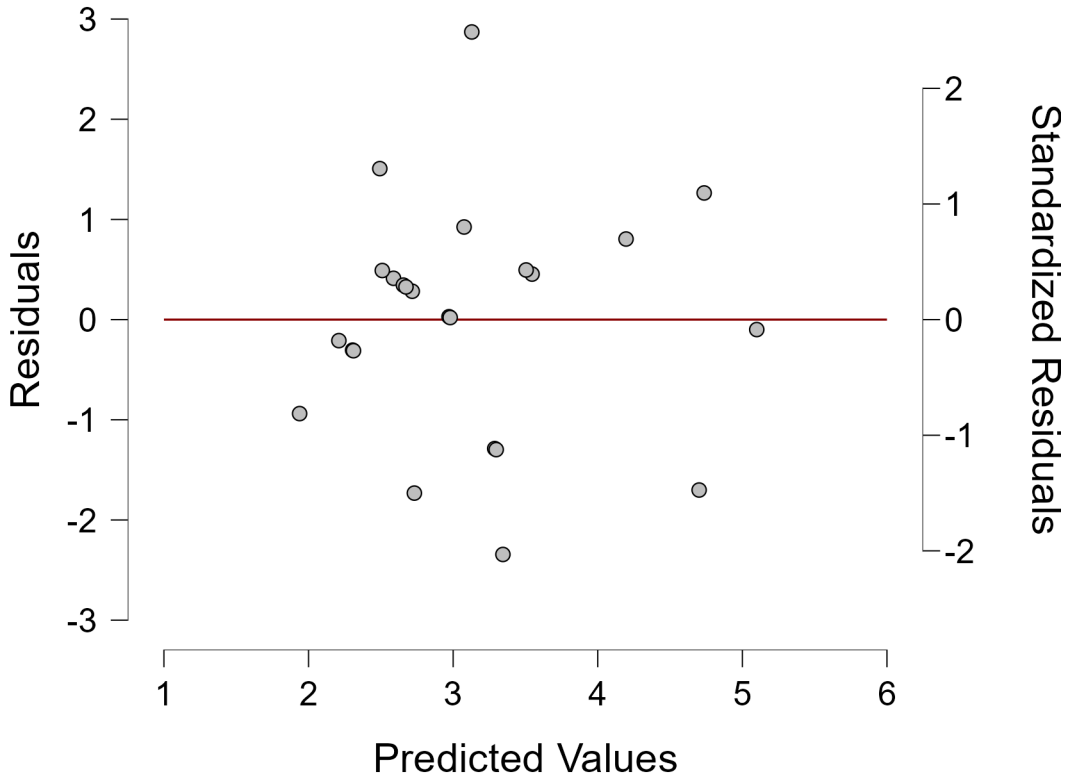
Model		log10_lux	Cover	Litter_cm	Tree_m
M_1	log10_lux	0.1242	0.002461	−0.01320	−0.007765
	Cover		9.601×10^{-4}	−0.001662	0.005750
	Litter_cm			0.02405	−0.01679
	Tree_m				0.1171

Note. The intercept model is omitted, as no meaningful information can be shown.

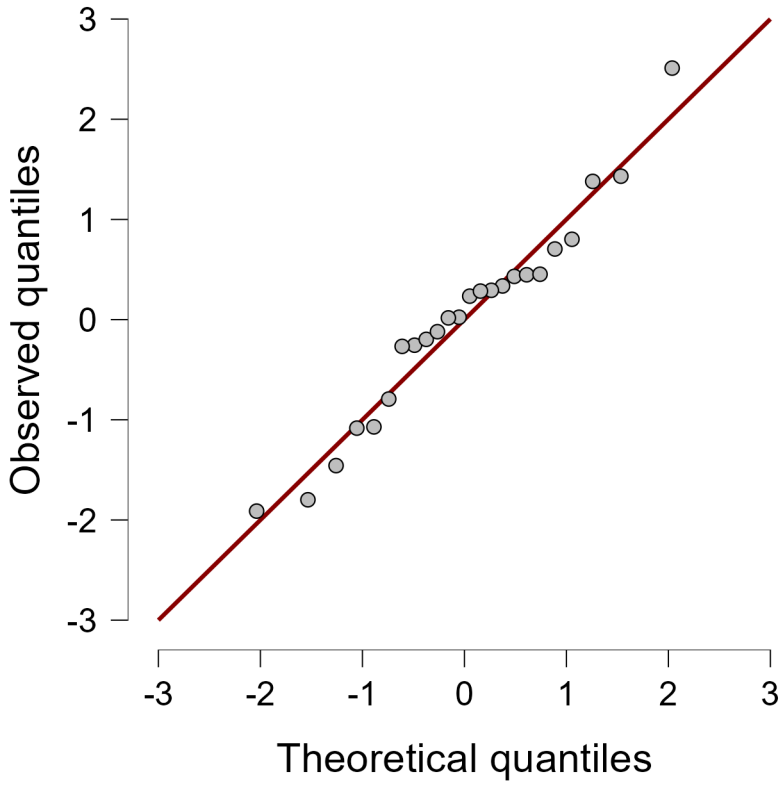
Residuals Statistics

	Minimum	Maximum	Mean	SD	N
Predicted Value	1.938	5.099	3.125	0.8338	24
Residual	−2.344	2.871	-1.758×10^{-16}	1.154	24
Std. Predicted Value	−1.423	2.368	2.498×10^{-16}	1.000	24
Std. Residual	−1.912	2.510	0.01636	1.031	24

Residuals vs. Predicted



Q-Q Plot Standardized Residuals



Linear Regression w/ Edge_m

Model Summary - Richness

Model	R	R^2	Adjusted R^2	RMSE	R^2 Change	df1	df2	p
M_0	0.000	0.000	0.000	1.424	0.000	0	23	
M_1	0.623	0.388	0.219	1.259	0.388	5	18	.089

Note. M_1 includes log10_lux, Cover, Litter_cm, Tree_m, Edge_m

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	18.11	5	3.622	2.286	.089
	Residual	28.51	18	1.584		
	Total	46.62	23			

Note. M_1 includes log10_lux, Cover, Litter_cm, Tree_m, Edge_m

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

							Collinearity Statistics	
Model		Unstandardized	Standard Error	Standardized	t	p	Tolerance	VIF
M_0	(Intercept)	3.125	0.2906		10.75	< .001		
M_1	(Intercept)	2.156	2.808		0.7676	.453		
	log10_lux	−0.5876	0.3505	−0.3355	−1.677	.111	0.848	1.179
	Cover	0.004736	0.03092	0.03596	0.1532	.880	0.616	1.622
	Litter_cm	0.1826	0.1586	0.2427	1.151	.265	0.764	1.309
	Tree_m	0.6563	0.3410	0.4461	1.925	.070	0.632	1.581
	Edge_m	−0.05102	0.04410	−0.2280	−1.157	.262	0.875	1.143

Coefficients Covariance Matrix

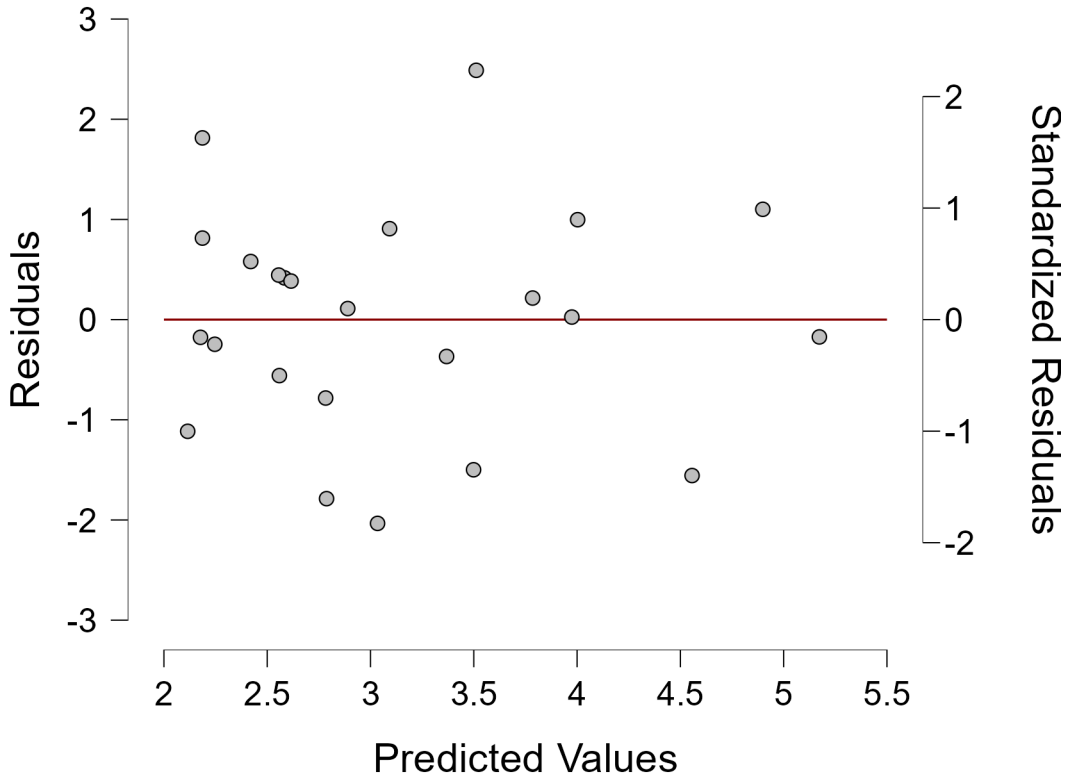
Model		log10_lux	Cover	Litter_cm	Tree_m	Edge_m
M_1	log10_lux	0.1228	0.002519	−0.01185	−0.006632	0.001273
	Cover		9.558×10^{-4}	−0.001495	0.005771	1.555×10^{-4}
	Litter_cm			0.02515	−0.01515	0.001721
	Tree_m				0.1163	0.001523
	Edge_m					0.001945

Note. The intercept model is omitted, as no meaningful information can be shown.

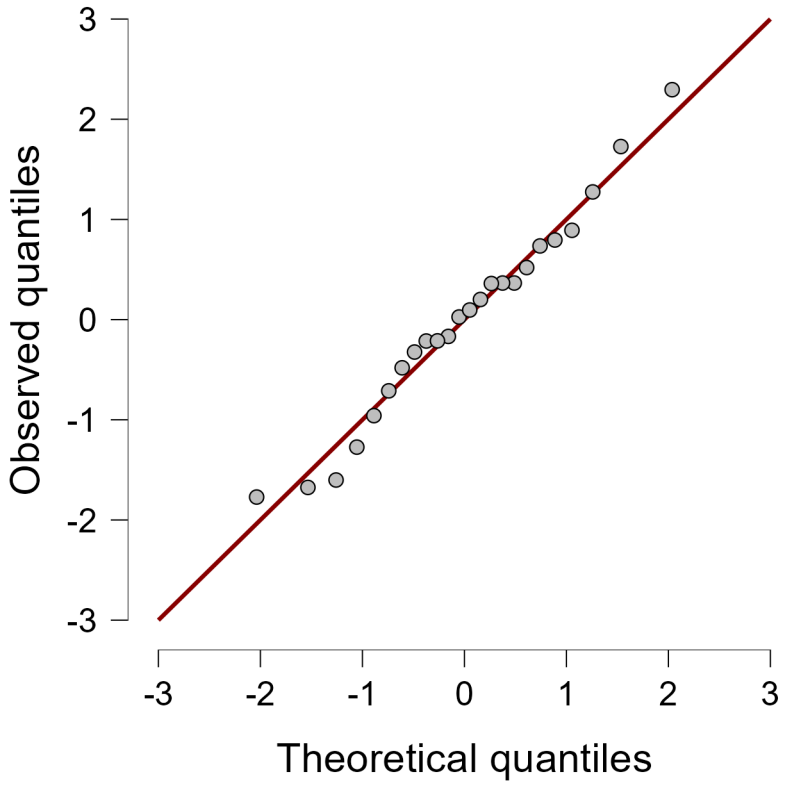
Residuals Statistics

	Minimum	Maximum	Mean	SD	N
Predicted Value	2.115	5.173	3.125	0.8874	24
Residual	−2.034	2.488	-6.014×10^{-17}	1.113	24
Std. Predicted Value	−1.138	2.308	-2.521×10^{-16}	1.000	24
Std. Residual	−1.771	2.295	0.01092	1.033	24

Residuals vs. Predicted



Q-Q Plot Standardized Residuals



Linear Regression w/ log10_Edge_m & w/o %cover

Model Summary - Richness

Model	R	R^2	Adjusted R^2	RMSE	R^2 Change	df1	df2	p
M_0	0.000	0.000	0.000	1.424	0.000	0	23	
M_1	0.655	0.429	0.309	1.183	0.429	4	19	.025

Note. M_1 includes log10_lux, Litter_cm, Tree_m, log10_Edge_m

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	20.02	4	5.005	3.575	.025
	Residual	26.60	19	1.400		
	Total	46.63	23			

Note. M_1 includes log10_lux, Litter_cm, Tree_m, log10_Edge_m

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

							Collinearity Statistics	
Model		Unstandardized	Standard Error	Standardized	t	p	Tolerance	VIF
M_0	(Intercept)	3.125	0.2906		10.75	< .001		
M_1	(Intercept)	3.398	1.037		3.277	.004		
	log10_lux	−0.6760	0.3251	−0.3860	−2.080	.051	0.872	1.147
	Litter_cm	0.1607	0.1437	0.2137	1.118	.277	0.822	1.216
	Tree_m	0.5449	0.2740	0.3704	1.989	.061	0.866	1.155
	log10_Edge_m	−1.236	0.7172	−0.3418	−1.724	.101	0.764	1.309

Coefficients Covariance Matrix

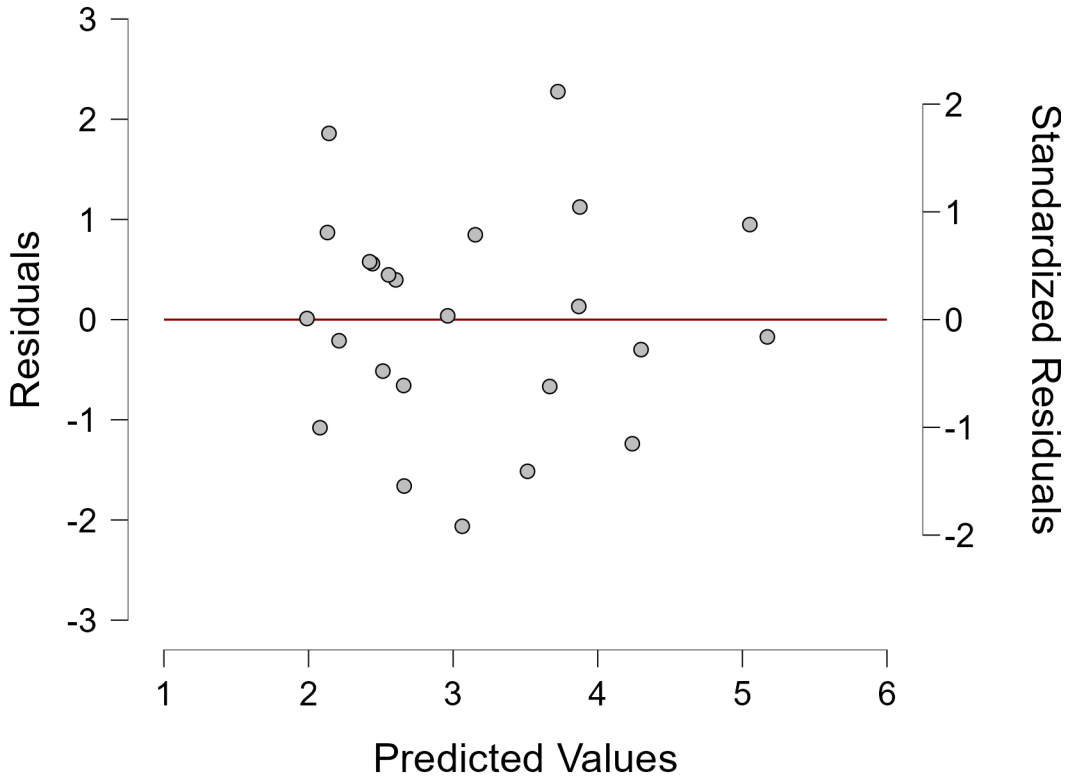
Model		log10_lux	Litter_cm	Tree_m	log10_Edge_m
M_1	log10_lux	0.1057	−0.005024	−0.01625	0.04126
	Litter_cm		0.02066	−0.003213	0.03422
	Tree_m			0.07509	0.04098
	log10_Edge_m				0.5143

Note. The intercept model is omitted, as no meaningful information can be shown.

Residuals Statistics

	Minimum	Maximum	Mean	SD	N
Predicted Value	1.989	5.173	3.125	0.9330	24
Residual	−2.063	2.275	-1.064×10^{-16}	1.075	24
Std. Predicted Value	−1.217	2.195	2.220×10^{-16}	1.000	24
Std. Residual	−1.858	2.155	-4.464×10^{-4}	1.019	24

Residuals vs. Predicted



Q-Q Plot Standardized Residuals

